

A cross-sectional study on social determinants of mental health during COVID-19 among adults in California

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Abstract

Aim: To explore the impact of structural and intermediary social determinants of health (SDoH) on Californian adults' mental health during the early phase of the COVID-19 pandemic.

Design: This cross-sectional study used data from the 2020 cycle of the California Health Interview Survey, the largest US state-level population health survey.

Methods: Descriptive statistics and logistic regression were used to analyse the data. Using a general social determinant of health framework, we operationalized different survey questions to measure structural and intermediary determinants of mental health.

Results: Mental health during the early phase of COVID-19 among adults in California was associated with age, gender, health conditions, delayed care, employment status (loss of job or reduced income) and discrimination. People in higher social strata were more likely to have better mental health for many of these factors.

Conclusion: This study supports the assertion that material circumstances (such as employment status) and discrimination are associated with experiencing mental health issues among adults in California during COVID-19. Racism is a public health issue, and as nurses, addressing racism is critical. In addition, much work is needed to address SDoH to improve health outcomes, especially among marginalized populations.

Impact: This study addressed the knowledge gap concerning the social determinants of mental health among Californian adults during the early phase of the COVID-19 pandemic. Those who had reduced income and those who lost their jobs during the COVID-19 pandemic were 46% and 56%, respectively, more likely to report mental health problems. Those who experienced discrimination in healthcare were 304% more likely to report mental health issues. This research will increase the understanding of the social determinants of health, particularly for those with chronic illnesses and mental health issues during the COVID-19 pandemic.

Patient or Public Contribution: No patient or public contribution, as we used an existing US state dataset. However, California Health Interview Survey is the largest state health survey in the United States and interviews more than 20,000 households each year representing the health care needs of Californians.

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KEYWORDS

CHIS, chronic diseases, COVID-19, mental health, pandemic, SDOH, type 2 diabetes, well-being

1 | INTRODUCTION

Social determinants of health (SDoH) are non-medical factors that influence overall health outcomes (World Health Organization [WHO], 2022a). These are conditions under which people are born, grow, work, live and age, including factors such as governance, macroeconomics, material circumstances and social policies that affect individual health conditions (Ratcliff, 2017; WHO, 2022a). Studies have shown that social factors significantly influence people's health risk exposure, susceptibility to diseases and disease development and outcomes (Cockerham et al., 2017; Madavanakadu Devassy et al., 2020). In addition, SDOH drives many deep-rooted inequities, particularly among historically marginalized populations.

SDoH can contribute to the development of mental health issues and chronic physical diseases such as type 2 diabetes, cardiovascular and pulmonary diseases and many other health conditions (Cockerham et al., 2017). As chronic diseases and mental health conditions are the most common incapacitating health issues experienced by many people, these problems, when they co-occur, pose self-management challenges leading to higher mortality rates (Momen et al., 2022; Scott et al., 2016). People who suffer from mental health and chronic disease conditions receive lesser care than people without mental disorders, and mental health patients are rarely referred to a mental health specialist as part of their treatment plan (Daré et al., 2019; Geissler & Zeber, 2020). Unfortunately, many studies examine the impact of SDOH on physical health but often do not include how the various factors of SDOH impacts the mental health of individuals (Cleveland et al., 2023; Cockerham et al., 2017).

2 | BACKGROUND

2.1 | COVID-19 and mental health

Since the outbreak of the SARS-CoV-2 (COVID-19) pandemic, the lives of people in the United States and around the world have changed significantly. The WHO (2022b) reported a 25% increase in the prevalence of depression and anxiety disorders in 2020 and 2021. The COVID-19 pandemic has forced people into isolation and brought with it job losses, lowered incomes, anxiety, fear, sorrow and grieving (Ornell et al., 2021). These stressors have been reported to increase the incidence of mental health issues and exacerbate preexisting psychiatric disorders (Gobbi et al., 2020; Ornell et al., 2021). A recent study reported that depressive, anxiety and post-traumatic stress symptoms were common in individuals after an acute phase of COVID-19, regardless of any previous psychiatric

diagnosis (Ornell et al., 2021). In addition, inequity in mental health treatment among minoritized groups is a significant problem due to structural racism (Mensah et al., 2021). Structural racism reinforces power hierarchies within a system and limits access to mental health resources (Mensah et al., 2021).

Although studies have shown how the experiences of individuals during COVID-19 exacerbated mental health issues, a limited number of studies examine how structural and intermediary determinants of health, such as changes in socioeconomic position and health system experiences during COVID-19, have impacted mental health. This is significant because the emotional dimension of an individual's health is often overlooked. As Gallo et al. (2021) explains, suboptimal self-care, excess disability and mortality are all consequences of poor mental health. Therefore, it is critical to understand the pathways in which various SDOH can lead to disparities in mental health and well-being between population groups and inequities in mental health care. As nurses and clinicians, we need to acknowledge and understand power hierarchies and how it impacts the mental health of individuals and develop an antiracist lens when providing mental health care to our patients and communities.

2.2 | Conceptual framework

We used the SDOH framework described by Jayasinghe (2015) and the WHO (2022a) to ground our study. In this conceptual framework, we posit that health inequity and well-being are associated with two broad types of SDOH: structural and intermediary factors. It demonstrates how structural determinants such as socioeconomic position, social class, gender, racism, education and intermediary determinants, which include material circumstances and health systems, play a role in determining health outcomes.

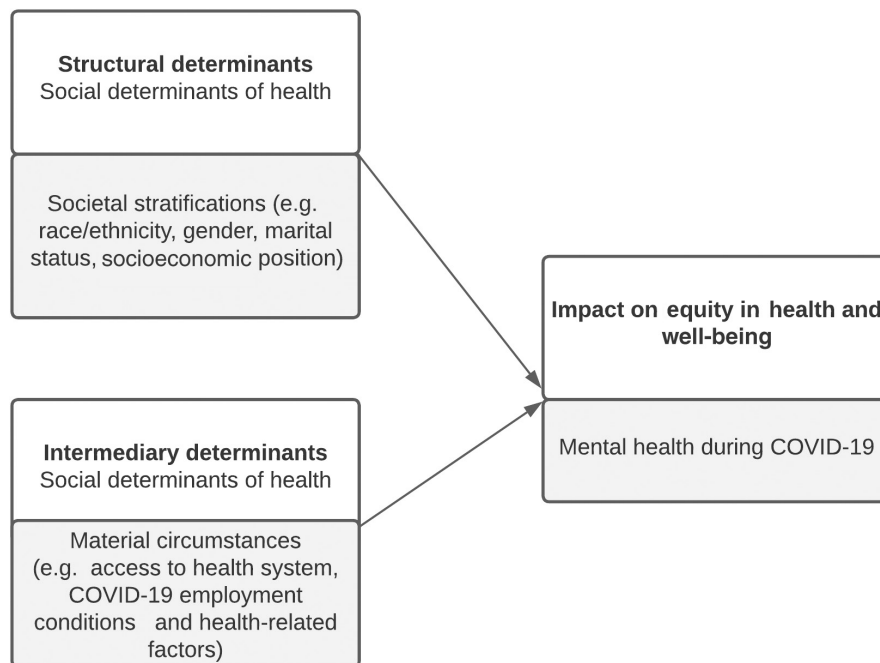
The operationalization of the various concepts applied in this study (Figure 1) is explained in the Measurements subsection.

3 | THE STUDY

3.1 | Aims

The aim of this study was to examine how SDOH, namely structural and intermediary determinants of health, has had an impact on individuals' mental health during the early phase of the COVID-19 pandemic. We explored how socioeconomic position, gender, ethnicity (racism), education, number of chronic diseases and material circumstances related to COVID-19 (access to health care and employment) impacted mental health during COVID-19.

FIGURE 1 Conceptual framework.



3.2 | Design

Data for this cross-sectional analysis came from the 2020 cycle of the California Health Interview Survey (CHIS), the largest US state-level population health survey (California Health Interview Survey, 2021). The CHIS covers various health topics, including health status, health conditions, mental health, health behaviours, employment and access to and use of health care.

3.3 | Participants

CHIS interviews are conducted with one randomly selected adult (18 years old and older) from each household using a secure access code. A probability sampling method was used, combined with population weights, to ensure the sample is representative of the non-institutionalized adult population of California at the county and state geographic levels (CHIS, 2021). This study used the CHIS 2021 adult sample ($N = 21,949$).

3.4 | Measurements

3.4.1 | Outcome variable: Impact on health equity in health/well-being variable (mental health)

We measured health and well-being using the concept of mental health. A cumulative index was used to operationalize the concept of mental health; specifically, participants were asked three questions related to depression and mental health disability care. A composite variable (a 'yes' response to any of the three survey questions) was then used for the health and well-being variable. See Table S1 for more information.

3.4.2 | Independent variable: Structural determinants of health

Our measures of structural determinants of health included structures rooted in institutions and processes that generate social stratification based on socioeconomic position, race/ethnicity, marital status and age. These structures reflect hierarchical systems that grant power and prestige to those at the higher end of the social scale.

Gender/ethnicity/age variables

We operationalized gender and ethnicity variables by using the following data found in the CHIS dataset: self-reported gender (male, female), race (Latino, American Indian/Native American, Asian, African American, White, Other/Multiple races) and self-reported age. We recoded age into five categories (18–34, 35–44, 45–54, 55–65, >65) instead of the 10 categories provided in the dataset. We used the survey question 'Were you treated unfairly due to race?' to measure the experience of discrimination or racism in healthcare.

3.4.3 | Independent variable: Intermediary determinants of health

We operationalized intermediary determinants using the concept of material circumstances (Jayasinghe, 2015), primarily focusing on access to health care and employment. We included specific survey questions about workplace experiences, including if respondents lost their job due to COVID-19, experienced reduced job hours or income due to COVID-19 or had difficulty paying for necessities. We also included two survey questions about their ability to work from home and if they were essential workers during the pandemic. All these questions had a binary response of 'yes' or 'no.'

As material circumstances also refer to health factors, we included questions regarding respondents' health status (Likert scale from *excellent*, *good*, *fair* to *poor*), presence of chronic disease/s ('yes' or 'no'), delayed or have not received medical care in the last 12 months ('yes' or 'no'), a preventative visit in the last 12 months ('yes' or 'no'), visiting the emergency room in the last 12 months ('yes' or 'no') and the time since they were last seen by a healthcare provider (categorized by 'within the past year,' '1-2 years ago,' '2-5 years ago,' '>5 years,' 'never').

3.5 | Data collection

CHIS is a mixed-mode (web and telephone) survey conducted every 2 years (CHIS, 2021). This study used the survey as it contained data from the sampling year 2020 related to COVID-19 and mental health (reported on 7 June 2021). UCLA Center for Health Policy Research staff addressed missing values in the CHIS data files through multiple imputations. A detailed methodology of the CHIS survey can be found on the CHIS website (<https://healthpolicy.ucla.edu/chis/Pages/default.aspx>).

3.6 | Ethical considerations

The CHIS data is available to the public. We did not use identifiable data; therefore, the analysis did not involve human subjects and did not require IRB review.

3.7 | Data analysis

We reported descriptive and inferential statistics for demographic variables in the study by calculating means, SD, frequency, the odds ratio, 95% CIs and level of statistical significance for each logistic regression, with mental health as our outcome variable and the various concepts that we operationalized to measure structural and intermediary determinants of health as our independent variables. In addition, given the high likelihood that mental health has potential confounders, we adjusted for all possible confounders (e.g. sociodemographic variables) apriori. These variables were selected because prior studies have shown that they might be associated with the likelihood of an individual experiencing mental health issues. All statistical analyses were performed using Stata, version 17 (Stata Corp.) and accounted for the complex survey design of the CHIS data, which included sampling weights.

3.8 | Validity and reliability

High-quality data from CHIS accurately reflect California's household population. CHIS uses the Total Survey Error data quality goals, a methodology that addresses many threats to survey quality.

4 | RESULTS

Table 1 presents the unweighted and weighted distributions of sociodemographic characteristics included in the study. For weighted sample characteristics, there was an almost equal representation between males (50.0%) and females (49.1%). The mean age of participants was 45.3 years old (SE=.03). Half of the respondents were married (50.5%), and the majority reported having good to excellent health (85.2%). About 15% of the respondents had chronic diseases, including asthma, diabetes and hypertension.

TABLE 1 Unweighted and weighted sample characteristics (Unweighted N = 21,949).

Variable	Unweighted N (%)	Weighted %
Sex		
Male	9575 (43.6)	49.1
Female	12,374 (56.38)	50.9
Race		
Latino/Latinx American	2328 (10.6)	23.5
Two or more races/other	2244 (10.2)	16.8
American-Indian/Alaska Native	147 (0.67)	0.7
Asian American	2755 (12.6)	13.5
Black/African American	778 (3.5)	5.7
White	13,687 (62.4)	39.8
Marital status		
Married	11,935 (54.4)	50.5
Divorced/widowed/separated/other	6255 (28.5)	24.2
Single never married	3759 (17.1)	24.4
General health condition		
Excellent	4545 (20.7)	20.5
Very good	8470 (38.6)	33.9
Good	6288 (28.7)	30.8
Fair	2190 (9.9)	12.7
Poor	456 (2.1)	2.1
Has/have chronic disease(s)		
None	17,994 (82.0)	84.8
Yes (types below)	3955 (18.0)	15.2
Diabetes (Type 1 or Type 2)	2317 (10.6)	10.5
HTN	8565 (39.9)	32.8
Asthma	3646 (16.2)	16.1
Heart disease	2127 (9.7)	6.5
Experienced mental health issues in the last 12 months	6938 (31.6)	36.2
Age	M = 53.1 (SD 17.6)	M = 45.3 (SE = 0.03)

Abbreviations: M, mean, SD, standard deviation, SE, standard error.

Over a quarter of the respondents experienced mental health issues in the last 12 months.

4.1 | Logistic regression results

Table 2 displays the unadjusted and adjusted logistic regression analyses examining the association of selected structural and intermediary determinants of health on mental health. We found that gender, age, racism/discrimination, socioeconomic position during COVID-19, health conditions and delayed care were all significant predictors of reporting or experiencing mental health issues.

Controlling for all other variables, the analysis indicated that female participants were 1.46 times more likely than males to experience mental health problems ($p = .04$, 95% CI [1.14, 1.87]). Those who reported being in excellent health were 0.63 times less likely to experience mental health problems than those in very good health ($p = .005$, 95% CI [0.46, 0.86]). However, those in fair health were 2.04 times more likely ($p = .003$, 95% CI [1.27, 3.26]), and those in good health were 1.66 times more likely ($p < .001$, 95% CI [1.23, 2.22]) to report mental health problems compared to those who had very good health.

Those who reported experiencing a delay or having not received medical care were 1.79 times more likely to have experienced mental health problems ($p < .001$, 95% CI [1.29, 2.48]). In addition, participants who lost their jobs due to the COVID-19 pandemic were 1.56 times more likely to report mental health problems ($p = .037$, 95% CI [1.03, 2.37]); in the same manner, participants who reported reduced income or job hours due to COVID-19 were 1.46 times more likely to report mental health problems ($p = .011$, 95% CI [1.09, 1.96]). Finally, participants who reported being treated unfairly due to their race in a healthcare system were 4.04 times more likely to report or experience mental health issues ($p = .018$, 95% CI [1.28, 12.80]).

5 | DISCUSSION

This study found that the impact on mental health during COVID-19 among California adults was associated with structural and intermediary determinants of health. For example, age, health status, delayed care, employment status (loss of job or reduced income) and discrimination based on race all contributed to increased mental health burdens.

For many of these factors, people who were more likely to be in a higher social stratum (such as those in excellent health vs. those in fair health; those employed vs. those unemployed; those treated fairly vs. those treated unfairly) were more likely to be in an environment that results in better mental health.

Those who reported experiencing delays in getting medical care or not receiving it were 79% more likely to experience mental health problems. Delayed treatment can increase morbidity and mortality, including developing a mental illness or worsening mental health conditions and physical comorbidities (Czeisler et al., 2020). Such

delays can also worsen treatment outcomes and decrease the likelihood of long-term recovery (Nguyen et al., 2019; Spivak et al., 2018). A prior study has also shown that temporary medical care access and delivery disruptions during COVID-19 impacted physical and emotional health (Czeisler et al., 2020). Therefore, as the COVID-19 pandemic continues and in preparation for future pandemics, it is crucial, as nurses, that we create and target care delivery strategies and communication initiatives that encourage people to seek timely care safely.

We also found that participants who lost their jobs and those who reported reduced income or job hours during the COVID-19 pandemic were, respectively, 56% and 46% more likely to report mental health problems. These findings demonstrate that factors such as loss of a job and reduced hours or income exacerbated mental health issues during the pandemic. A recent study among adults in the United States indicated that national estimates of self-reported anxiety disorder symptoms were nearly three times higher during COVID-19 in the second quarter of 2020 compared with the second quarter of 2019 (Guerin et al., 2021). Also, the prevalence of depressive disorder symptoms was nearly four times higher (Guerin et al., 2021). The profound negative impact on the mental health of people losing jobs and income is well established (de Miquel et al., 2022), and this study reinforces that pattern during a period when COVID-19 led to a historic loss of jobs and income in the United States (Gould & Kassa, 2021).

Furthermore, participants who reported being treated unfairly due to race are 304% more likely to report or experience mental health issues. Discrimination is a significant problem among members of ethnic minority groups in the United States. The COVID-19 pandemic uncovered racial health and social inequities that stem from a legacy of legally sanctioned racism and contemporary forms of racial discrimination against minoritized groups (Chae et al., 2021). For example, Chae et al. (2021) found that Asian Americans reported leaving their houses cautiously and only when necessary to avoid discrimination and violence. In the same light, Black/African Americans were also afraid when leaving their homes during the COVID-19 pandemic for fear of discrimination or being attacked. Discrimination can cause added stress, leading to mental health problems in these populations and preventing them from engaging with healthcare institutions and receiving the care needed for chronic health conditions (Cockerham et al., 2017). Zhou et al., (2021), also found that racial discrimination is associated with increased mental health problems.

Interestingly, we found in this analysis that Black/African Americans were 62% less likely to experience mental health issues than whites. Similarly, a recent study found that during the COVID-19 pandemic, Black Americans exhibited comparatively low levels of depression and anxiety compared to other groups (Owens & Saw, 2021). In contrast, we found no significant differences in mental health issues among non-Black racial and ethnic groups (e.g. Asian Americans, Native Americans, American Indians and Latinx) compared to Whites. Given that Black Americans experience several types of mental health stressors more than other racial and ethnic

TABLE 2 Unadjusted logistic (uOR) and adjusted logistic (aOR) regressions of structural and intermediary determinants of health on mental health.

Variable	uOR	SE	p-value	95% CI		aOR	SE	p-value	95% CI	
Sex (reference: male)										
Female	1.22	0.05	<.001**	1.12	1.33	1.46	0.18	.004*	1.14	1.87
Age (ref: >65 years old)										
18–34 years old	3.86	0.26	<.001**	3.37	4.41	3.53	0.85	<.001**	2.19	5.71
35–44 years old	2.29	0.15	<.001**	2.00	2.62	1.97	0.42	.002*	1.28	3.02
45–64 years old	1.87	0.122	<.001**	1.64	2.13	1.76	0.41	.019*	1.10	2.80
Race (ref: White)										
Latino/Latinx American	1.33	0.09	<.001**	1.16	1.53	0.97	0.13	.853	0.74	1.28
Two or more races/other	1.39	0.09	<.001**	1.21	1.58	0.88	0.20	.572	0.57	1.37
American Indian/Alaskan Native	0.66	0.19	.0144*	0.38	1.16	0.49	0.48	.47	0.07	3.50
Asian American	1.15	0.07	.037*	1.00	1.30	1.22	0.18	.198	0.90	1.64
Black/African American	0.84	0.09	.101	0.69	1.03	0.38	0.13	.006*	0.20	0.75
Treated unfairly due to race (ref: no)										
Yes	2.80	0.48	<.001**	1.99	3.92	4.04	2.34	.018*	1.28	12.80
Marital status (ref: married)										
Other/Separated/divorced	1.50	0.09	<.001**	1.33	1.69	1.18	0.21	.350	0.83	1.67
Never married	2.34	0.15	<.001**	2.07	2.65	1.28	0.20	.115	0.94	1.75
Health condition (ref: very good)										
Excellent	0.63	0.04	<.001***	0.56	0.72	0.63	0.10	.005**	0.46	0.86
Good	1.35	0.07	<.001***	1.21	1.50	1.66	0.25	<.001**	1.23	2.22
Fair	1.58	0.12	<.001***	1.36	1.84	2.04	0.48	.003**	1.27	3.26
Poor	2.50	0.02	<.001***	0.48	0.56	2.96	3.86	.406	0.22	39.6
Have chronic disease (ref: yes)										
No	1.12	0.05	.011*	1.02	1.22	0.82	0.11	.152	0.62	1.08
Time since last seen doctor (ref: 1–2 years ago)										
Within the past year	0.88	0.13	.411	0.66	1.19	0.99	0.27	.959	0.57	1.70
2–5 years	0.99	0.14	.970	0.75	1.31	1.04	0.17	.825	0.75	1.44
>5 years	0.84	0.15	.330	0.59	1.20	0.82	0.15	.273	0.58	1.17
Never	1.48	0.51	.263	0.74	2.95	1.53	0.65	.322	0.65	3.58
Delayed/did not get medical care in the last 12 months (ref: no)										
Yes	2.82	0.15	<.001***	2.54	3.14	1.79	0.29	<.001**	1.29	2.48
Had preventative visit in the last 12 months (ref: no)										
Yes	0.78	0.04	<.001***	0.71	0.86	0.92	0.27	.779	0.52	1.64
Visited Emergency Room in the last 12 months (ref: No)										
Yes	1.46	0.10	<.001***	1.28	1.66	1.46	0.53	.307	0.70	3.01
Lost job due to COVID-19 (ref: no)										
Yes	1.75	0.12	<.001***	1.52	2.01	1.56	0.33	.037*	1.03	2.37
Reduced job hours or income due to COVID-19 (ref: no)										
Yes	1.52	0.09	<.001***	1.35	1.71	1.46	0.22	.011*	1.09	1.96
Worked from home due to COVID-19 (ref: no)										
Yes	1.33	0.08	<.001***	1.19	1.49	1.09	0.16	.530	0.82	1.46
Essential worker (ref: no)										
Yes	1.16	0.06	.003**	1.05	1.29	1.10	0.16	.524	0.82	1.46

TABLE 2 (Continued)

Variable	uOR	SE	p-value	95% CI		aOR	SE	p-value	95% CI	
Difficulties paying for basic necessities (ref: no)										
Yes	2.39	0.24	<.001***	1.96	2.91	1.10	0.26	.671	0.69	1.75

Note: Outcome variable is an index of three mental health questions related to depression and mental health disability care. For the adjusted model, we controlled for sex, age, race, marital status, health condition, presence of chronic disease, time since last seen a doctor, delayed medical care, previous visit with doctor in 12 months, visited ER in 12 months, lost job due to COVID, reduced job hours or income, working from home, essential worker, difficulties paying for basic necessities and treated unfairly due to race.

* $p < .05$; ** $p < .01$; *** $p < .001$.

groups, this finding seems paradoxical. However, research has revealed that Black/African Americans are particularly susceptible to the prevalence and impacts of stigma related to mental illness, which frequently discourages them from getting care for their mental health (Rivera et al., 2021). While the study showed that Black/African Americans were less likely to experience mental health issues, we think this is due to the stigma associated with mental illness, thus, underreporting of experiences of such events. It is crucial to note that mental illness, such as depression, usually goes undiagnosed and untreated in the Black/African American population with chronic diseases such as type 2 diabetes (Ajuwon & Insel, 2022). Further research is needed to examine the role of stigma in mental health disparities among Black/African Americans to develop strategies to address this phenomenon.

Nursing can play a pivotal role in addressing the different factors that contribute to mental health issues experienced by individuals. Understanding the various effects of SDoH on mental health during COVID-19 can help nurses develop evidence-based strategies that can be used for education, nursing praxis and advocacy to mitigate the effects of discrimination, racism and patriarchal systems that disadvantage historically marginalized groups. Mensah et al. (2021) highlighted several actionable recommendations in the mental health arena to achieve mental health equity that nurses can act upon: (1) investigation of the power differential between those in positions of authority and vulnerable or ignored positions that exist throughout society, including in our clinical praxis; (2) conduct individual and institutional reflection and analyses of racial inequities; and (3) application of an antiracist lens to advocate for mental health care equity.

5.1 | Limitations

As the CHIS is a state-level population health dataset, one limitation of this analysis is that it limits extending the external validity of our results to the rest of the United States. However, the CHIS is modelled after the National Health Interview Survey (NHIS), and many questions that could be appropriately answered in the NHIS can also be answered by the CHIS. In addition, the CHIS is administered much more frequently than the NHIS, so its data is generally more recent, and it oversamples racial/ethnic minorities, which is well suited for looking at issues related to health inequities related to race/ethnicity. Also, we were limited to operationalizing the various concepts included in this study by the questions in the survey. However, our

using of a cumulative index for mental health, for instance, assured that we captured any respondents who self-reported experiencing mental health issues. Finally, as this was an observational cross-sectional study, we cannot definitively infer causality.

6 | CONCLUSION

We found a significant association between people experiencing mental health issues during the early phase of the COVID-19 and economic factors. In addition, delayed care and discrimination were also associated with mental health problems. These findings highlight the importance of instituting social or governmental policies that help limit work losses and maintain financial security, addressing issues related to discrimination and racism, removing the barrier of the stigma associated with seeking mental help and ensuring that we provide safe, efficient and timely care to all.

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CONFLICT OF INTEREST STATEMENT

The authors do not have any disclosures, conflicts of interest, or financial interests to report.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/jan.15803>.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the California Health Interview Survey (CHIS) at <https://healthpolicy.ucla.edu/chis/Pages/default.aspx>.

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